23 **Ans: B** (N86 P1 Q17)

At pd V = 300 V, charge stored Q = CV = (20 x 10^{-6} F) x (300 V)

Using Q = It where I = a **constant current** of 10 mA,

$$t = (20 \times 10^{-6}\ \text{F}) \times (300\ \text{V}) \div (10 \times 10^{-3}\ \text{A}) = 0.60\ \text{s}$$

24 **Ans: D**